



Faculty of Engineering

School of Electrical Engineering and Telecommunications

ELEC 1111 – Topic 8

AC Analysis II

Dr. Inmaculada (Inma) Tomeo-Reyes

Senior Lecturer

School of Electrical Engineering and Telecommunications, UNSW

Topic 8 Content

Corresponds to Chapter 10
of your textbook

This lecture covers:

- Steady-state analysis of AC circuits in frequency domain.
 - Nodal analysis.
 - Mesh analysis.
- Application of superposition principle in frequency domain.
 - Circuits operating at a **single** frequency.
 - Circuits operating at **different** frequencies.
- Application of source transformation in frequency domain.
- Application of Thevenin's and Norton's theorems in AC circuits.
- AC Op Amp circuits.

Addressed learning outcomes:

- *LO1*: Systematically analyse DC and AC electric circuits by deriving and solving equations using circuit laws and theorems.
- *LO3*: Apply phasors and sinusoidal steady state analysis to AC circuits.
- *LO6*: Evaluate relevant information to design simple engineering systems that use electrical circuits.

Analysing AC circuits

- When a circuit is operated by a **sinusoidal source**, its **steady-state response** can be obtained by using **phasors**.
- Transforming the circuit to phasor/frequency domain makes the analysis much simpler as we would no longer require to solve differential equations.
- Analyzing AC circuits usually require three steps:
 1. **Transform** the circuit to **phasor/frequency domain**.
 2. **Solve** the problem using **circuit analysis techniques** (nodal or mesh analysis, superposition, source transformation, etc.).
 3. **Transform** the resulting phasors **back** to **time domain** (if required).

AC circuit analysis is performed in the **same manner** as **DC circuit analysis** except that **complex numbers** are involved.

Nodal analysis in AC circuits

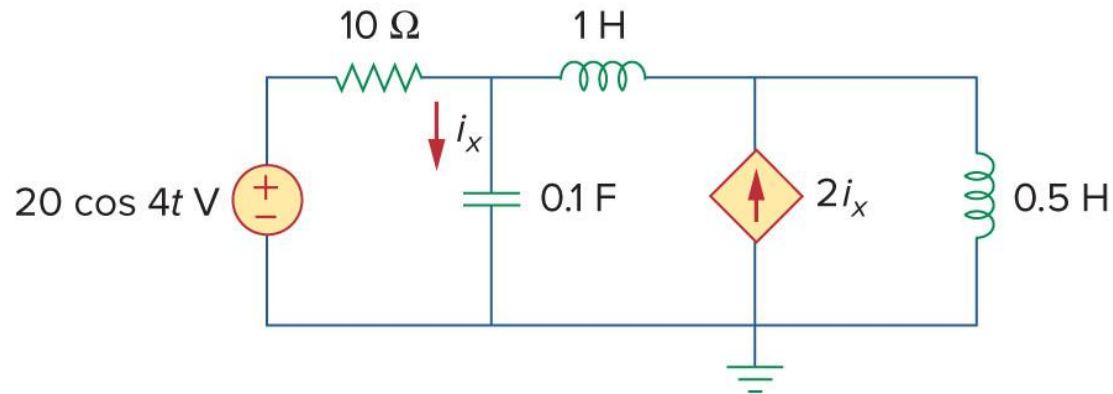
- We use **KCL** to write **nodal equations** in an AC circuits as we did in DC circuits.
 - The only difference is that **voltage phasors** at **each node** should be used.
- Always **convert a time-domain circuit to phasor/frequency domain** (if the circuit is not given in phasor domain) by calculating **all the impedances** of the circuit elements at the operating frequency and replacing the **sinusoids** (in cosine form) with their phasors.
- We deal with **voltage sources** in nodal analysis the same way as in DC:

1. Voltage source between non-reference node and reference node (i.e. ground):
 - Assign the **node voltage** to the **source voltage phasor**.
2. Voltage source between two non-reference nodes:
 - Form a **supernode**.
 - Apply KCL at the supernode using the already assigned phasor voltages at nodes inside supernode.
 - Write the extra equation relating node voltages inside supernode and the source voltage phasor.

Exercise 1

Find i_x in the circuit given below using nodal analysis.

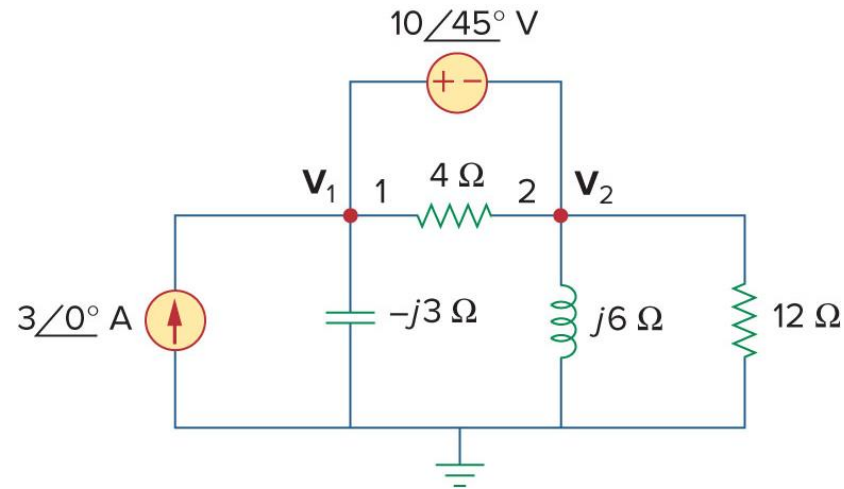
Solution in Exercises_solution_Topic_8.pdf



Exercise 2

Compute V_1 and V_2 in the circuit below using nodal analysis.

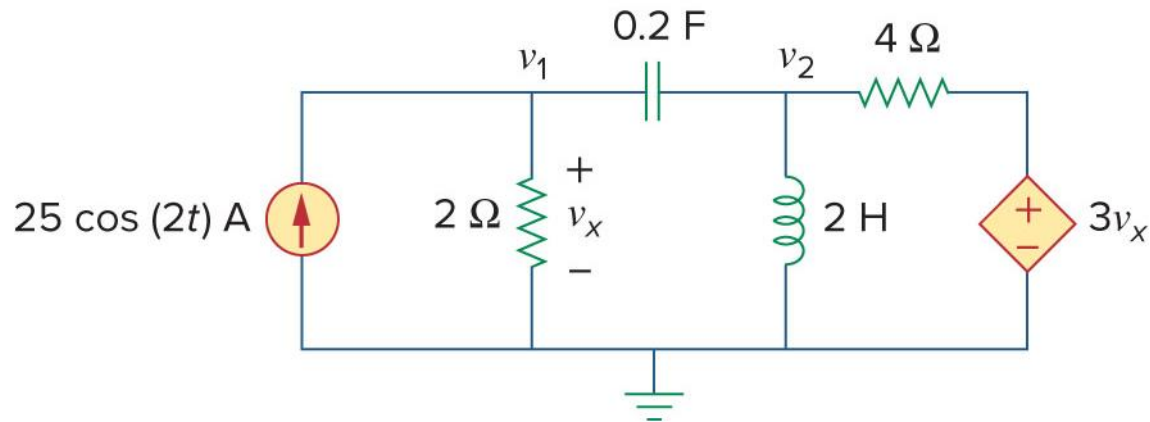
Solution in Exercises_solution_Topic_8.pdf



Exercise 3

Compute v_1 and v_2 in the circuit below using nodal analysis

- For practice!
- Answer: $v_1(t) = 28.31 \cos(2t + 60.01^\circ) \text{ V}$
 $v_2(t) = 82.56 \cos(2t + 57.12^\circ) \text{ V}$



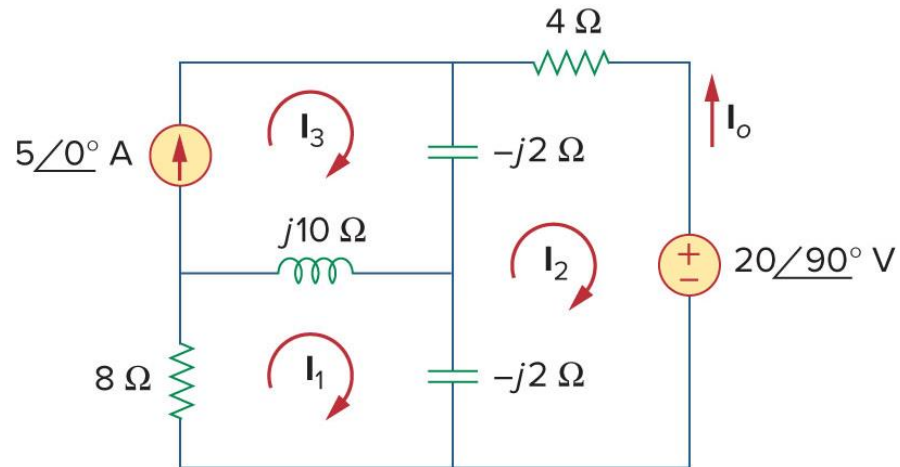
Mesh analysis in AC circuits

- We use **KVL** to write **mesh equations** in an AC circuits as we did in DC circuits.
 - The only difference is that **current phasors** in **each mesh** should be used.
- Always **convert a time-domain circuit to phasor/frequency domain** (if the circuit is not given in phasor domain) by calculating **all the impedances** of the circuit elements at the operating frequency and replacing the **sinusoids** (in cosine form) with their phasors.
- We deal with **current sources** in mesh analysis the same way as in DC:
 1. Current source belonging to ONLY one mesh/loop:
 - The **mesh current** in that mesh/loop is **equal** to the **source current phasor**.
 2. Current source shared between two meshes:
 - Form a **supermesh**.
 - Apply KVL in the supermesh using the already assigned mesh current phasors in the supermesh.
 - Write the extra equation relating mesh currents in the supermesh and the source current phasor.

Exercise 4

Determine current I_o in the circuit below using mesh analysis.

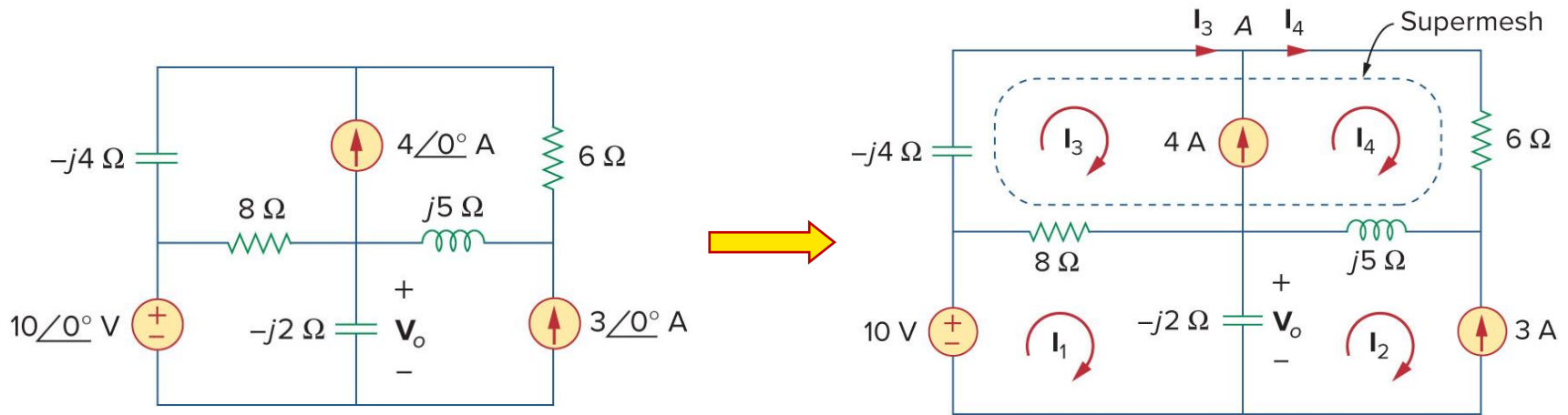
Solution in Exercises_solution_Topic_8.pdf



Exercise 5

Find the voltage V_o in the circuit below using mesh analysis.

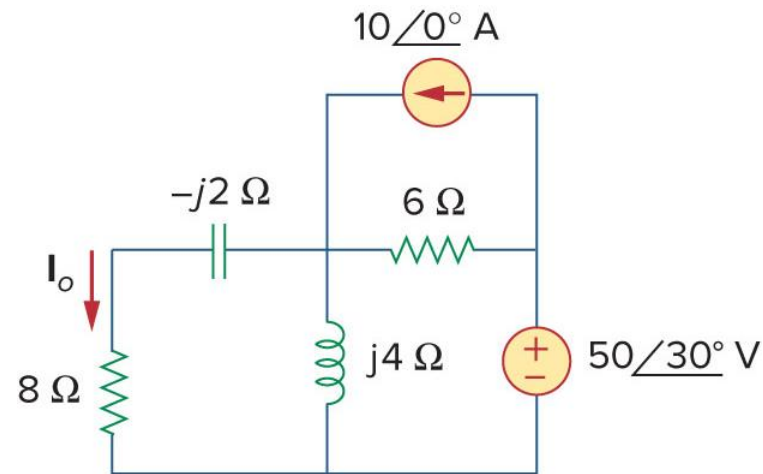
Solution in Exercises_solution_Topic_8.pdf



Exercise 6

Calculate I_o in the circuit below using mesh analysis.

- For practice!
- Answer: $I_o = 5.97 \angle 65.45^\circ \text{ A}$



Superposition theorem in AC circuits

- Since AC circuits are **linear**, it is also possible to apply the principle of **superposition** in AC circuits with **multiple sources**.
- This becomes **particularly important** if the circuit has **sources** operating at **different frequencies**.
- The complication is that **each source** must have its **own frequency-domain** equivalent circuit because:
 - Impedances depend on frequency which means each element has a different impedance at different frequency ($Z = R(\omega) + jX(\omega) \Omega$).
 - **Phasor voltages and phasor currents** resulting from each **different-frequency source CANNOT be added** to each other **in frequency domain**, instead they all must be converted back to time domain before being added.

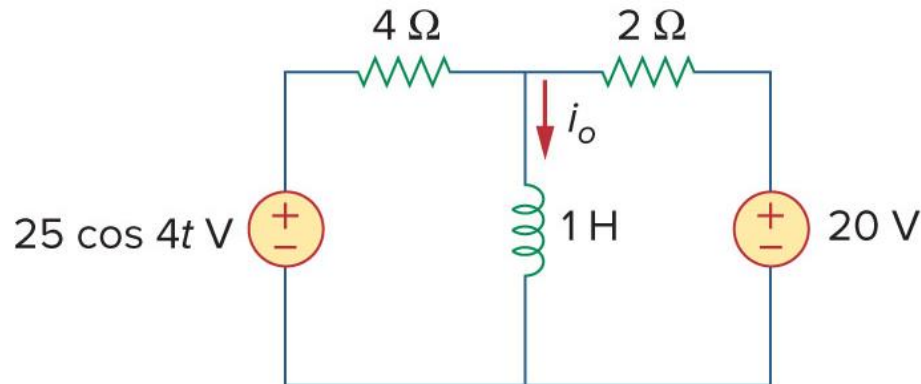
Check the frequency of all sources in the circuit before applying **superposition** to make sure whether you need to **recalculate** the **impedances** for each frequency-domain equivalent circuit.

Exercise 7

Find i_o in the circuit shown below using superposition.

Note: This is an example of the use of superposition with sources operating at different frequencies (AC source at 4 rad/s and DC source at 0 rad/s).

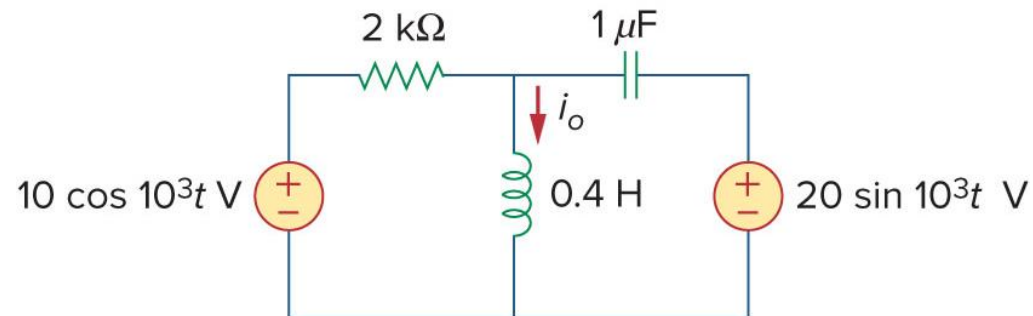
Solution in Exercises_solution_Topic_8.pdf



Exercise 8

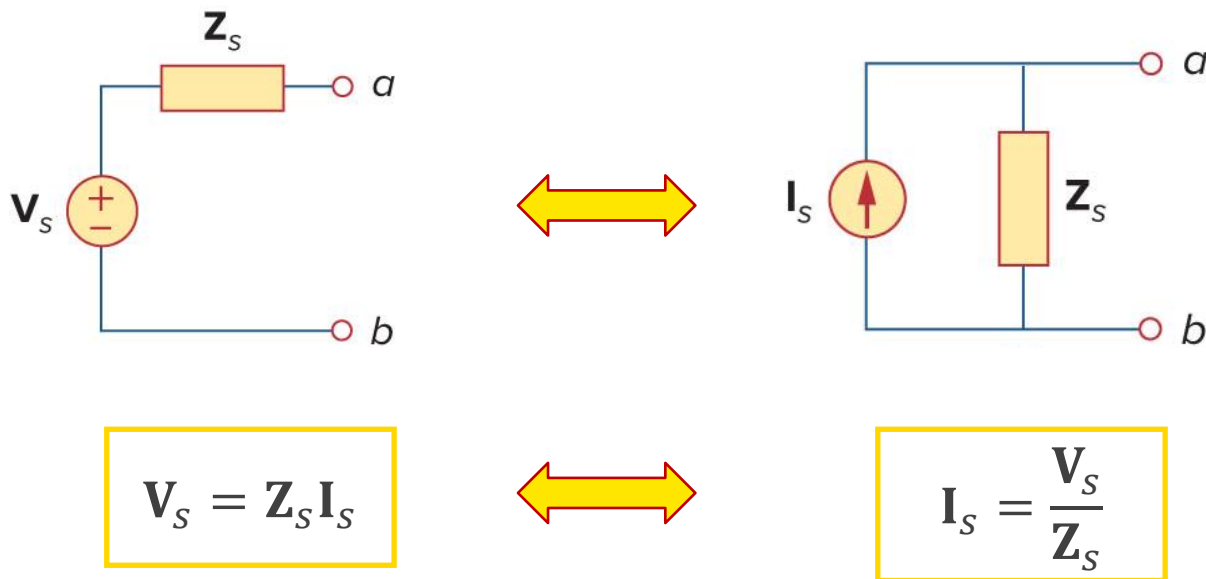
Use superposition to calculate i_o in the following circuit.

- For practice!
- Answer: $i_o = 39.5 \cos(10^3 t - 18.43^\circ) \text{ mA}$



Source transformation

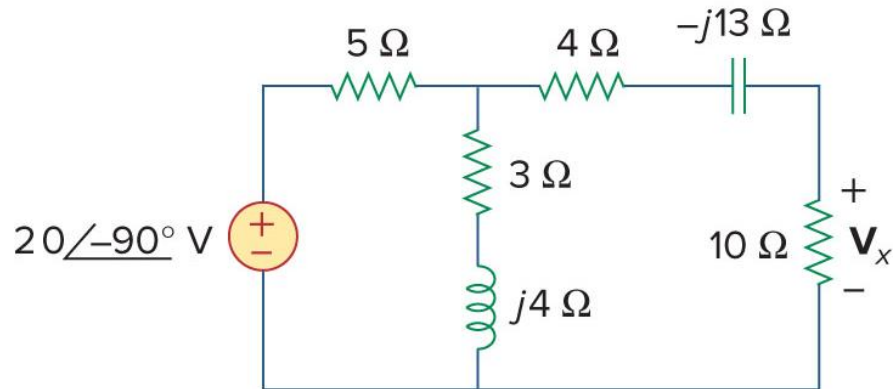
Source transformation in the frequency domain involves transforming a voltage source in series with an impedance to a current source in parallel with an impedance or vice versa.



Exercise 9

Calculate V_x in the circuit shown below using source transformation.

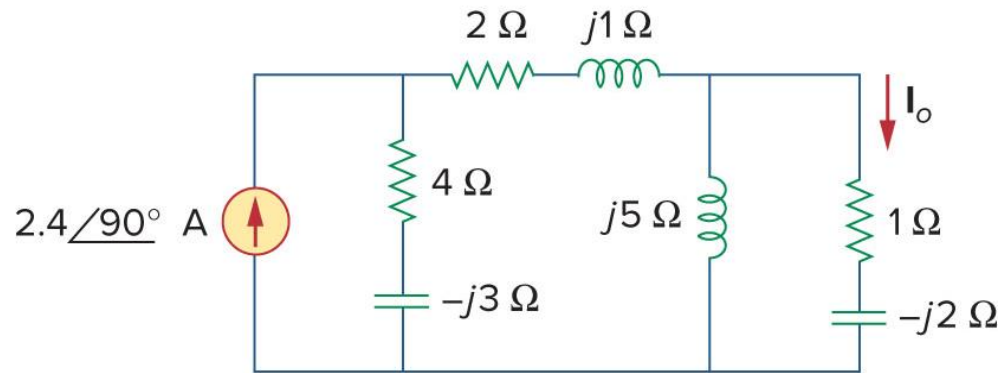
Solution in Exercises_solution_Topic_8.pdf



Exercise 10

Use source transformation to find I_o in the following circuit.

- For practice!
- Answer: $I_o = 1.97 \angle 99.46^\circ \text{ A}$



Thevenin's and Norton's Theorems

- Both Thevenin and Norton's theorems are applied to linear AC circuits the **same way** as in DC linear circuits.
 - The only **difference** is the fact that the calculated values will be **complex**.
- The two equivalent circuits are **related** through **source transformation**.

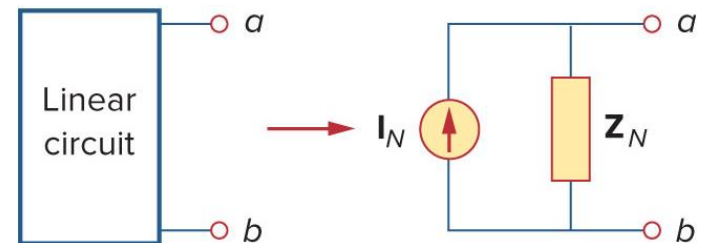
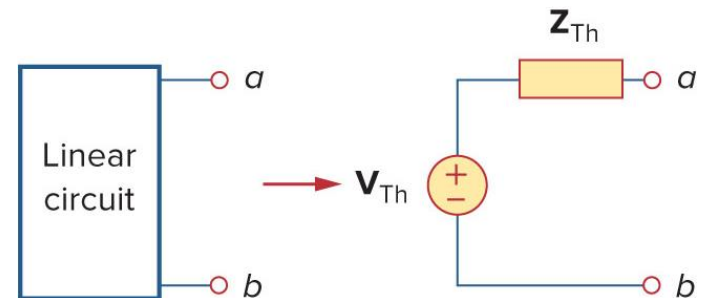
$$V_{Th} = Z_N I_N$$



$$Z_{Th} = Z_N$$

$$I_N = \frac{V_{Th}}{Z_{Th}}$$

V_{Th} : Open-circuit voltage across terminals a - b .
 I_N : Short-circuit current through terminals a - b .
 $Z_{Th} = Z_N$: Equivalent or input impedance seen from terminals a - b .



Thevenin's and Norton's Theorems

- Finding the $\mathbf{Z}_{Th} = \mathbf{Z}_N = \mathbf{Z}_{eq} = \mathbf{Z}_{in}$ is **the same** as in DC circuits:
 - Turn off all independent sources and calculate equivalent impedance from terminals (not possible if there is any dependent source).
 - Use definition of Thevenin resistance:

$$\mathbf{Z}_{Th} = \frac{\mathbf{V}_{Th}}{\mathbf{I}_N}$$

- Connect an external source (only possibility if there are ONLY dependent sources).

- Connect a **constant voltage source** in phasor form $\mathbf{V}_o = 1\angle 0^\circ \text{ V}$ across the terminals (turn off independent sources if any).
- Find the phasor current $\mathbf{I}_o$ through the external voltage source
- $\mathbf{Z}_{eq} = \frac{\mathbf{V}_o}{\mathbf{I}_o} = \frac{1\angle 0^\circ}{\mathbf{I}_o}$ (Passive sign convention)

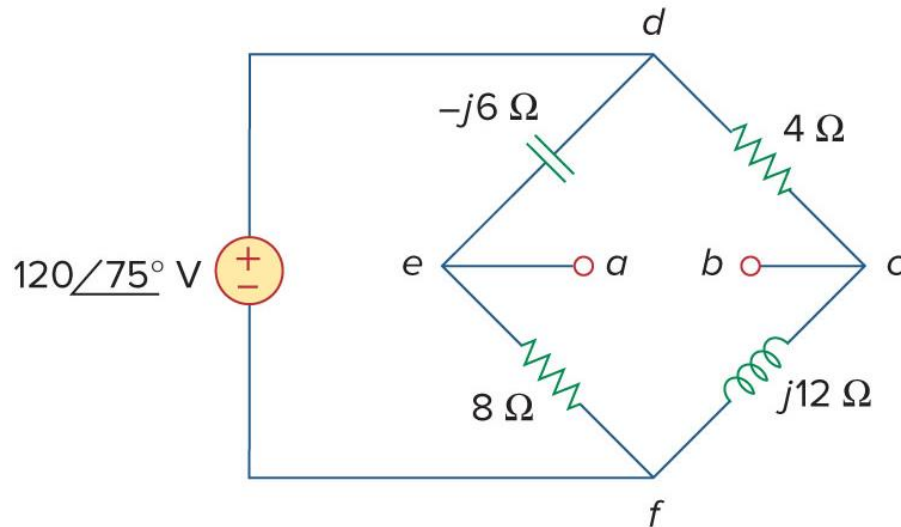
- Connect a constant current source in phasor form $\mathbf{I}_o = 1\angle 0^\circ \text{ A}$ across the terminals (turn off independent sources if any).
- Find the phasor voltage $\mathbf{V}_o$ across the external current source
- $\mathbf{Z}_{eq} = \frac{\mathbf{V}_o}{\mathbf{I}_o} = \frac{\mathbf{V}_o}{1\angle 0^\circ}$ (Passive sign convention)

The **value** of the constant external source **does not** have to be **one**. It is just for simplicity.

Exercise 11

Obtain the Thevenin equivalent circuit at the terminals a - b in the circuit below.

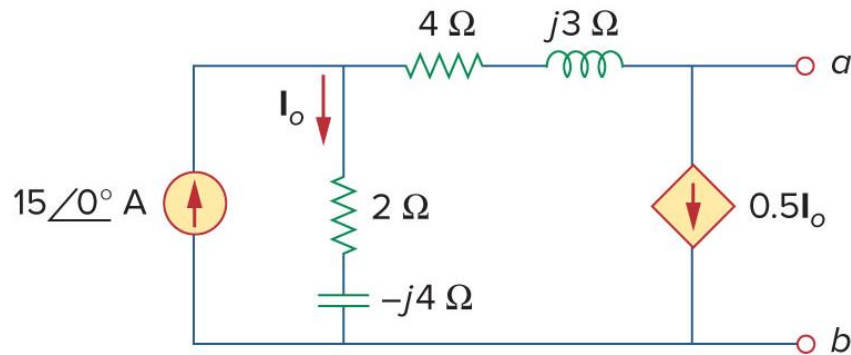
Solution in Exercises_solution_Topic_8.pdf



Exercise 12

Find the Thevenin equivalent circuit as seen from the terminals a - b in the circuit below.

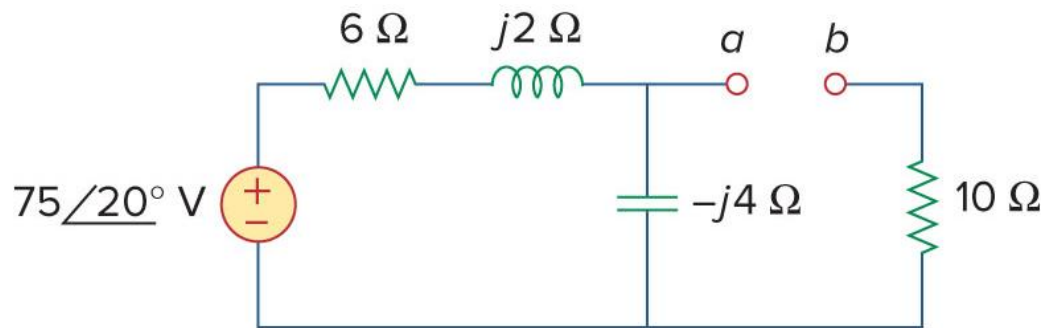
Solution in Exercises_solution_Topic_8.pdf



Exercise 13

Obtain the Norton equivalent circuit at the terminals a - b in the circuit below.

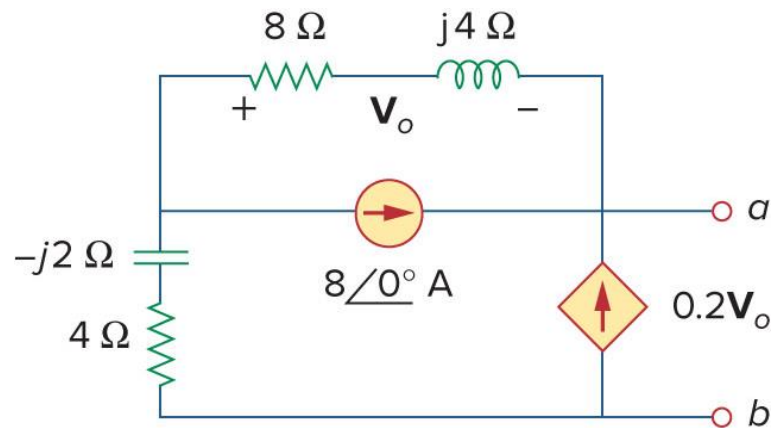
- For practice!
- Answer: $\mathbf{Z}_N = 12.4 - j3.2 = 12.8\angle -14.47^\circ \Omega$, $\mathbf{I}_N = 3.703\angle -37.1^\circ \text{ A}$



Exercise 14

Determine the Thevenin equivalent circuit at the terminals a - b in the circuit below.

- For practice!
- Answer: $\mathbf{Z}_{Th} = 4.47 \angle -7.64^\circ = 4.43 - j0.594 \Omega$, $\mathbf{V}_{Th} = 11.763 \angle 72.9^\circ \text{ V}$



Analysing AC Op Amp circuits

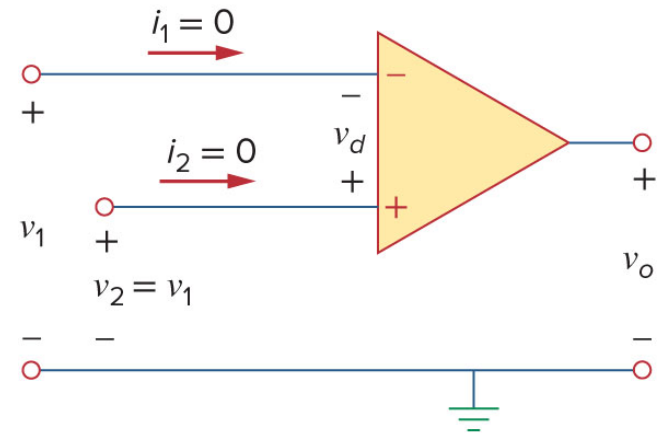
- If an Op Amp is connected to an AC source, we can use analysis in the phasor domain to find the steady state voltages and currents in the circuit.
- We use **nodal analysis** to solve AC Op Amp circuits in phasor domain.
- We consider **ideal Op Amp** and the following rules:

1. The **input phasor currents** to Op Amps terminal are **zero**.

$$I_1 = I_2 = 0$$

2. The input **phasor voltages** at **inverting** and **non-inverting** terminals are **equal** (only if there is a **negative feedback** connection from Op Amp's output to its inverting input).

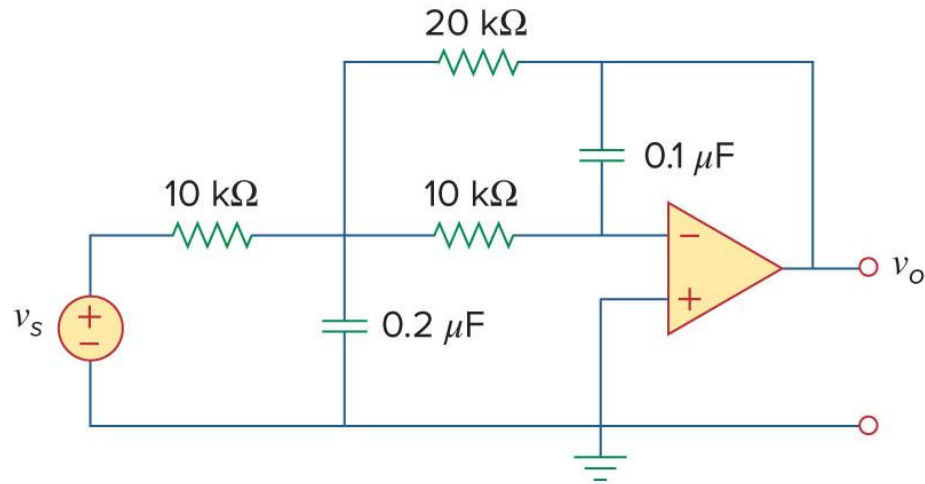
$$V_1 = V_2$$



Exercise 15

Determine $v_o(t)$ for the Op Amp circuit shown below if $v_s(t) = 3 \cos(1000t)$ V.

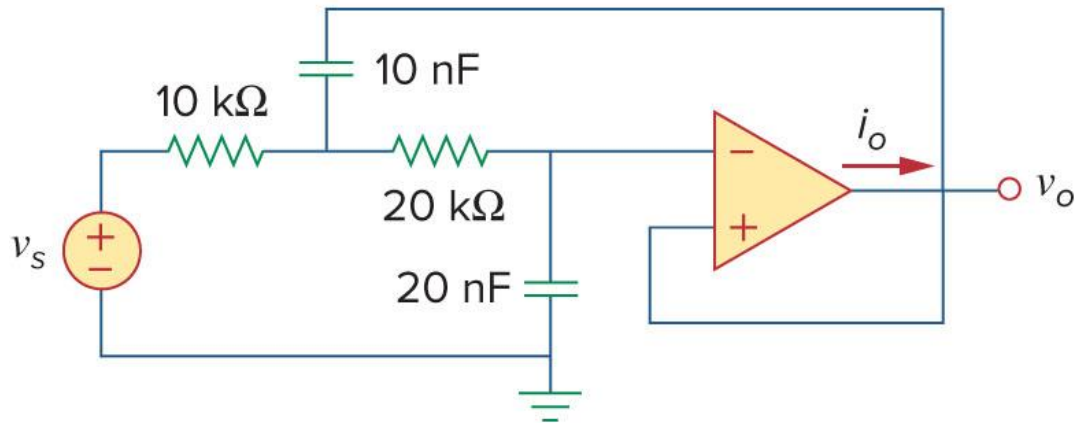
Solution in Exercises_solution_Topic_8.pdf



Exercise 16

Find v_o and i_o in the Op Amp circuit shown below if $v_s(t) = 12 \cos(5000t)$ V

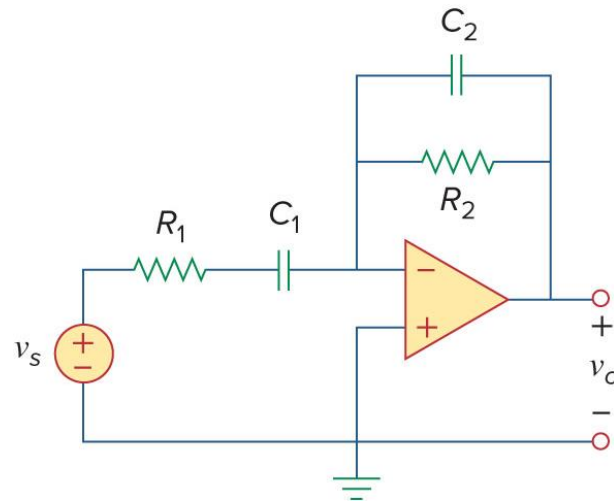
- For practice!
- Answer: $v_o(t) = 4 \cos(5000t - 90^\circ)$ V
 $i_o(t) = 0.4 \cos(5000t - 90^\circ)$ mA



Exercise 17

Compute the closed loop gain and the phase shift for the circuit below if $R_1 = R_2 = 10 \text{ K}\Omega$, $C_1 = 2 \mu\text{F}$, $C_2 = 1 \mu\text{F}$ and $\omega = 200 \text{ rad/s}$.

Solution in Exercises_solution_Topic_8.pdf

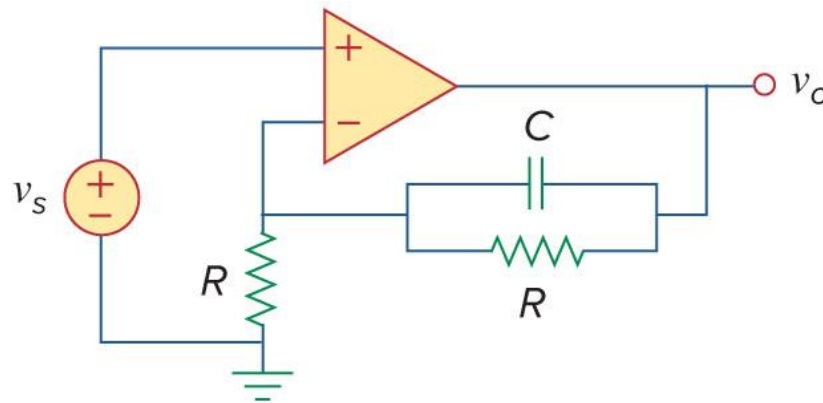


Exercise 18

Obtain the closed loop gain and the phase shift for the circuit below if $R = 10\text{ K}\Omega$, $C = 1\text{ }\mu\text{F}$, and $\omega = 1000\text{ rad/s}$ (non-inverting amplifier).

- For practice!
- Answer: $\mathbf{G} = \frac{V_o}{V_s} = 1.0147\angle(-5.6^\circ)$

The gain is $|\mathbf{G}| = 1.0147$ and the phase shift is $\angle\mathbf{G} = -5.6^\circ$.



Questions?

